

Calculus 2 - Homework 2 (Problems)

1.

Evaluate $\int 1/(x(x-1)^2) dx$.

2.

Evaluate $\int \tan^2(x) \sec^2(x) dx$.

3.

Evaluate $\int \sqrt{9-x^2} dx$.

4.

Evaluate the definite integral $\int_0^{\pi/4} \tan(x) \sec^2(x) dx$.

5.

Evaluate $\int e^x \sin(x) dx$.

6.

Evaluate the definite integral $\int_0^1 \arctan(x) dx$.

7.

Evaluate $\int (x+3)/((x-1)(x+2)) dx$.

8.

Determine whether $\int_1^{\infty} 1/x dx$ converges or diverges.

9.

Use the Root Test to determine whether $\sum_{n=1}^{\infty} (n/(2n+1))^n$ converges or diverges.

10.

For $\int_1^2 (1/x) dx$, use the Trapezoidal Rule error bound $|E_T| \leq K(b-a)^3 / (12n^2)$ to find the smallest number of subintervals n that guarantees an error less than 0.001.